

MOKOHINAU, New Zealand

WMO: 930690

Lat: 35.903S

Lon: 175.115E

Elev: 197

StdP: 14.59

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	49.1	50.5	36.0	31.2	51.7	38.6	34.6	52.7	45.7	55.7	42.5	55.3	18.7	180	0.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.5	74.2	67.6	72.9	66.8	71.9	66.2	70.4	72.1	69.1	71.1	68.1	70.3	9.4	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.9	110.7	71.3	68.4	105.0	70.3	67.2	100.8	69.6	34.5	72.4	33.4	71.2	32.5	70.3	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
43.2	37.3	32.8	DB	46.8	76.7	1.9	1.7	45.4	77.9	44.3	79.0	43.3	79.9	41.9	81.2	
			WB	41.4	71.3	1.5	1.7	40.3	72.5	39.4	73.6	38.5	74.5	37.4	75.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.5	67.4	68.7	67.3	64.4	61.0	57.5	55.9	55.7	56.8	58.5	61.0	64.7
	DBStd	5.17	2.39	2.19	2.28	2.56	2.83	2.74	2.42	2.33	2.30	2.22	2.35	2.45
	HDD50	2	0	0	0	0	0	0	1	1	0	0	0	0
	HDD65	1583	6	2	7	40	126	226	281	290	245	203	122	35
	CDD50	4206	540	523	535	432	341	224	184	176	205	262	329	455
	CDD65	312	81	105	77	21	2	0	0	0	0	0	1	25
	CDH74	44	12	27	3	0	0	0	0	0	0	0	0	2
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	16.4	14.7	14.7	15.2	15.4	16.8	17.7	18.7	17.2	17.7	16.8	16.7	15.1
(15) Precipitation	PrecAvg	52.0	3.0	3.3	4.0	4.4	4.6	5.5	6.0	5.6	4.4	3.8	3.5	3.9
	PrecMax	70.7	9.0	10.1	8.5	8.9	10.8	9.8	15.5	10.8	11.7	6.9	8.6	9.3
	PrecMin	34.0	0.9	0.9	1.4	1.5	1.3	2.2	1.3	2.2	2.2	1.2	1.5	1.0
	PrecStd	9.4	1.9	2.1	1.9	1.9	2.1	2.0	2.9	1.7	1.8	1.4	1.5	2.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.5	76.5	74.3	71.3	68.1	64.5	62.7	61.6	63.4	65.1	69.2	72.9
		MCWB	68.0	69.2	67.1	66.8	64.4	62.3	59.9	58.6	58.9	59.4	62.3	65.8
	2%	DB	73.8	74.7	72.7	69.9	66.6	63.3	61.2	60.5	62.0	63.8	67.2	70.9
		MCWB	67.1	68.2	66.4	65.4	63.1	60.5	58.0	57.4	58.1	58.8	61.3	64.6
	5%	DB	72.5	73.4	71.6	68.8	65.7	62.2	60.2	59.8	61.0	62.8	65.9	69.6
		MCWB	66.4	67.3	65.9	64.6	62.2	59.0	56.8	56.5	57.5	58.3	60.6	63.7
	10%	DB	71.3	72.2	70.6	67.8	65.0	61.4	59.4	59.0	60.1	61.9	64.8	68.4
		MCWB	65.8	66.7	65.3	63.9	61.3	57.8	55.9	55.5	56.8	57.8	60.1	63.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.4	72.5	70.6	69.6	66.2	63.3	61.1	60.3	61.0	61.9	64.7	68.7
		MCDB	72.8	73.6	71.7	70.2	67.0	64.0	62.2	60.7	61.9	63.0	66.4	70.1
	2%	WB	69.6	71.1	69.0	67.4	64.7	61.6	59.2	59.1	59.7	60.8	63.5	67.1
		MCDB	71.8	72.6	70.9	68.5	65.9	62.8	60.4	59.9	60.9	62.2	65.2	68.8
	5%	WB	68.3	69.8	67.8	66.1	63.6	60.2	58.0	57.8	58.9	60.0	62.5	66.1
		MCDB	70.6	71.8	70.2	67.9	65.2	61.8	59.6	59.1	60.3	61.7	64.4	67.9
	10%	WB	67.3	68.4	66.7	64.8	62.3	58.8	56.9	56.4	58.0	59.1	61.5	65.0
		MCDB	69.9	70.8	69.5	67.2	64.5	60.9	58.9	58.3	59.7	61.3	63.9	67.1
(35) Mean Daily Temperature Range	5% DB	MDBR	6.8	6.5	5.8	5.0	4.4	4.1	4.2	4.5	5.2	5.9	6.6	6.7
		MCDBR	8.0	7.9	6.6	5.6	4.6	4.6	4.5	4.8	5.7	6.4	7.8	8.1
		MCWBR	4.7	4.5	4.4	4.2	4.2	4.7	4.5	4.3	4.6	4.8	5.0	5.0
	5% WB	MCDBR	6.1	6.0	5.6	4.6	4.0	4.2	4.2	4.1	4.8	5.2	6.2	6.0
		MCWBR	4.5	4.7	4.9	4.8	4.6	5.3	5.1	4.7	4.6	5.2	5.2	4.6
(40) Clear-Sky Solar Irradiance	taub		0.345	0.343	0.331	0.319	0.310	0.306	0.304	0.309	0.330	0.334	0.335	0.341
	taud		2.597	2.632	2.652	2.677	2.667	2.655	2.656	2.647	2.562	2.563	2.586	2.616
	Ebn at Noon		314	308	300	286	271	263	270	285	295	307	315	317
	Edh at Noon		33	30	28	24	21	20	21	24	30	32	33	32
(44) All-Sky Solar Radiation	RadAvg		2185	1885	1565	1137	808	656	718	984	1346	1747	2032	2097
	RadStd		123	131	103	75	50	43	42	56	68	105	120	175

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page